Current Epidemiology Reports (2020) 7:203–211
https://doi.org/10.1007/s40471-020-00245-2

EPIDEMIOLOGIC METHODS (P HOWARDS, SECTION EDITOR)

Difference-in-Difference in the Time of Cholera: a Gentle Introduction for Epidemiologists

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Accepted: 11 September 2020 / Published online: 23 September 2020

Abstract

Purpose of Review The goal of this article is to provide an introduction to the intuition behind the difference-in-difference method for epidemiologists. We focus on the theoretical aspects of this tool, including the types of questions for which difference-in-difference is appropriate, and what assumptions must hold for the results to be causally interpretable.

Recent Findings While currently under-utilized in epidemiologic research, the difference-in-difference method is a useful tool to examine effects of population level exposures, but relies on strong assumptions.

Summary We use the famous example of John Snow's investigation of the cause of cholera mortality in London to illustrate the difference-in-difference approach and corresponding assumptions. We conclude by arguing that this method deserves a second look from epidemiologists interested in asking causal questions about the impact of a population level exposure change on a population level outcome for the group that experienced the change.

Keywords Difference in difference · Change scores · Causal inference · John Snow

Introduction

The difference-in-difference method is one of the oldest tools in the epidemiology toolkit, first used in 1855 by John Snow in his analysis of the cause of cholera [1•, 2•, 3•]. Despite this, difference-in-difference is used relatively infrequently by epidemiologists today. In this review, we describe what the difference-in-difference method is, how and why the estimation procedure works for estimating causal effects, and what types of causal questions it is appropriate for. We propose a causal diagram (DAG) for difference-in-difference analyses and use Snow's investigation of London's cholera outbreak as a motivating example to outline the method.

This article is part of the Topical Collection on *Epidemiologic Methods*

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What Is the Difference-in-Difference Method and How Does It Work?

In its simplest form, the difference-in-difference method is a comparison of the change over time in a continuous outcome variable (such as the probability or rate of an outcome) in an exposed group versus the change over time in an unexposed or control group [3•, 4•]. This approach was pioneered by John Snow in 1855 to understand the causes of cholera mortality. Table 1 reproduces his data on cholera mortality rates by household water source.

In 1855 London, cholera was a common occurrence, but the cause of cholera was unknown. John Snow suspected that the infection was waterborne and designed a difference-in-difference study to test his hypothesis. In his investigation of the source of cholera, he began by comparing mortality from cholera across London and noticed that areas served by the Southwark and Vauxhall Company and the Lambeth Water Company experienced particularly high mortality during the 1849 outbreak [1•, 2•]. Both of these companies pumped water from the River Thames to households across the city. In many cases, the companies served neighboring houses, greatly reducing the possibility that any differences in rates of cholera deaths by water supply company could be explained by differences in neighborhood and socioeconomic factors. The intake stations for these two companies were likewise located on adjacent segments of the river and in both cases were downstream of the city's sewage outflow sites. However, in 1852, the Lambeth Water Company moved its water intake to a new site upstream of the sewage outflow [1•, 2•]. This provided a compelling 'natural experiment' for John Snow to test his theory that cholera was waterborne. For epidemiologists unfamiliar with the term, a 'natural experiment' is a situation in which a change in exposure status is believed to have occurred at random with respect to the counterfactual outcomes and one from which observational data can therefore be analyzed directly as if it were obtained from a randomized trial.

While John Snow wanted to obtain evidence about the cause of cholera in an *individual*, he in fact used the water company data to answer a causal question about *group level* infection: did the rate of cholera death decrease when the water intake from the Lambeth Company was moved upstream of the sewage outlet compared with *what it would have been if that water intake had remained downstream* (Box 1)? This question can be answered quantitatively by obtaining an estimate of the average causal effect *in the treated*—sometimes referred to as the average treatment effect in the treated or ATT [4•].

Box 1 The causal contrast of interest

	Definition	Application to cholera example
A causal contrast of interest appropriate for difference-in-difference methods	Average causal effect in the treated: The change in the outcome over the follow-up time *in the treated* had everyone been treated compared with The change in the outcome over the follow-up time *in the treated* had nobody been treated	The change in the rate of cholera death over 5 years *in households served by the Lambeth Water Company* had the water intake been moved upstream of the sewage outlet between 1849 and 1854 compared with The change in the rate of cholera death over 5 years *in households served by the Lambeth Water Company* had the water intake remained downstream between 1849 and 1854

To answer this question, Snow needed to estimate the counterfactual cholera death rate that would have been observed in the households served by the Lambeth Water Company if they had not moved their intake source. Because cholera death rates vary considerably over time, the cholera death rate in 1849 for households serviced by the Lambeth Company would not provide a valid estimate of this counterfactual rate. Snow noted, however, that the rate of cholera death among households served by the Southwark & Vauxhall Company increased by 118 deaths per 100,000 people from 1849 to 1854 and that since the original water sources were nearly identical, one might reasonably expect the same increase for households served by the Lambeth Company had the pump not been moved. Instead, Snow observed that the rate of cholera death decreased by 653 deaths per 100,000 in households served by the Lambeth Company during this period; thus, emerged the difference-in-difference approach. As shown in Table 1, Snow's causal question could be answered by comparing the change in cholera death rates over time among households served by the Lambeth Company with the change in cholera death rates over time among households served by the Southwark & Vauxhall Company.

We next describe in formal notation the causal estimand in which Snow was interested, the corresponding estimator and estimate, and the assumptions required to endow the resulting estimate with the desired causal interpretation.

Formalizing the Difference-in-Difference Estimation Procedure: Estimand, Estimator, and Estimate

The Causal Question

Before formalizing the causal estimand, it is important to take a moment to discuss the causal *question* of interest in a little more depth. Snow asked: did the rate of cholera death decrease when the water intake from the Lambeth Company was moved upstream of the sewage outlet compared with *what it would have been if that water intake had remained downstream*? In epidemiology, many of our causal questions of interest are related to individual-level biological mechanisms. Indeed, Snow's ultimate goal was to triangulate information in a way that allowed him to infer the causes of cholera within the individual. However, the difference-in-difference method is designed to answer causal questions posed at the group level. These types of questions include the effects of policies implemented at the population level. For example, Snow's study inferred the effect of the business decision to change the water source—a change which operated on the entire population of Lambeth Company customers.

A common concern among epidemiologists when conducting or reviewing studies of population level exposures is the potential for making the inferential error called the 'ecological fallacy'—that is, assuming that processes that are detected at the population level provide meaningful inference about processes at the individual level. Less well-recognized, but equally important, is the 'aggregation fallacy'. This fallacy reminds us that it is equally incorrect to assume that processes that are detected at the individual level will provide meaningful inference about processes at the population level [5]. The difference-in-difference method provides us with a tool that can be used to obtain unbiased answers to causal questions about population level processes.

Finally, it is important to be aware that the difference-in-difference method allows us to answer a specific subset of questions about a population: that is, what would have happened to the group that *experienced* the change, if they had contrary to fact not experienced the change. This causal question can provide some evidence to triangulate guesses about what might happen to the other group (the control group) if it were to in the future experience a similar change, but that is not the target of inference and can require additional assumptions.

The Causal Estimand

John Snow's causal question can be re-written as a formal causal estimand using counterfactual notation. First, let A represent the exposure of interest: $A = 1$ if the water source is the Lambeth Company and $A = 0$ if the water source is the Southwark & Vauxhall Company. Let $a = 1$ denote the intervention "move the water source upstream" and $a = 0$ denote the intervention "keep the water source downstream". Next, let Y represent the outcome of interest for an individual at the end of follow-up ($Y = 1$ if a person dies of cholera; $Y = 0$ if a person does not die of cholera). Let Y^a represent the counterfactual outcome, death from cholera, that a person would experience at the end of follow-up *if* they received water from source $A = a$. For example, $Y^{a=1}$ is the counterfactual outcome an individual would experience if they received water from an upstream water source. The counterfactual outcomes are defined for all individuals in the study sample, but as we shall see, we are interested only in a contrast of counterfactual outcomes among those individuals who were in fact served by the Lambeth Company. Let t represent time, where $t = 0$ at baseline and $t = 1$ at end of follow-up (5 years post-baseline). Y_t denotes the observed outcome at a given time point and $Y_{t=1} - Y_{t=0}$ is the 5-year change in the rate of cholera death. We use Δt to denote this 5-year change and Δt^a to denote the 5-year change in the counterfactual outcome, death from cholera. Finally, we let L denote common causes of $Y_{t=0}$ and Δt.

Now that we have described our notation, we can draw a causal diagram to depict the causal contrast of interest (Fig. 1).

Finally, we can write the causal estimand of interest using the notation described above:

$$\Theta_{\Delta t} = E\left[\Delta t^{a=1} | A = 1\right] - E\left[\Delta t^{a=0} | A = 1\right] \quad (1)$$

In words, the causal estimand is the 5-year change in the average rate of cholera death that would have been observed in households served by the Lambeth Company if the company had changed its water source to be upstream between 1849 and 1854 compared with the average rate of cholera that would have been observed in those same households at that same time if the Lambeth Company had not changed its water source to be upstream during that same time period (Box 2a).

Box 2 The causal estimand, estimator, and estimate of interest in the difference-in-difference method

	Definition	Application to cholera example
2a. Estimand	$\Theta_{\Delta t} = E[\Delta t^{a=1} \| A = 1] - E[\Delta t^{a=0} \mid A = 1]$ (1) where $E[\Delta t^{a=1} \|\| A = 1]$ is the average change in the outcome over follow-up that would have been observed among the exposed ($A = 1$) had everyone been exposed ($a = 1$) and where $E[\Delta t^{a=0} \mid A = 1]$ is the average change in the outcome over follow-up that would have been observed among the exposed ($A = 1$) had nobody been exposed ($a = 0$)	$E[\Delta t^{a=1} \| A = 1]$ is the average change in the rate of cholera death over 5 years that would have been observed for households served by the Lambeth company if the company had changed its water source to upstream between 1849 and 1854 $E[\Delta t^{a=0} \| A = 1]$ is the average change in the rate of cholera death over 5 years that would have been observed for households served by the Lambeth company if the company had not changed its water source to upstream between 1849 and 1854
2b. Estimator	$\Theta_{\Delta t} = E[\Delta t^{a=1} \| A = 1] - E[\Delta t^{a=0} \mid A = 1]$ (1) $= \left(E\left[Y^{a=1}_{t=1} \| A = 1\right] - E\left[Y^{a=1}_{t=0} \| A = 1\right]\right) - \left(E\left[Y^{a=0}_{t=1} \| A = 1\right] - E\left[Y^{a=0}_{t=0} \| A = 1\right]\right)$ $= (E[Y \| A = 1, T = 1] - E[Y \| A = 1, T = 0]) - (E[Y \| A = 0, T = 1] - E[Y \| A = 0, T = 0])$ (2)	$E[Y \| A = 1, T = 1]$ is the average cholera death rate in households served by the Lambeth Company in 1854 $E[Y \| A = 1, T = 0]$ is the average cholera death rate in households served by the Lambeth Company in 1849 $E[Y \| A = 0, T = 1]$ is the average cholera death rate in households served by the Southwalk & Vauxhall Company in 1854 $E[Y \| A = 0, T = 0]$ is the average cholera death rate in households served by the Southwalk & Vauxhall Company in 1849
2c. Estimate	$\widehat{\Theta}_{\Delta t} = (E[Y \| A = 1, T = 1] - E[Y \| A = 1, T = 0]) - (E[Y \| A = 0, T = 1] - E[Y \| A = 0, T = 0]) = (193.4 - 846.7) - (1466.1 - 1348.6) = -770.8$ (3)	Same

Table 1 John Snow's data on mortality from cholera in areas served by only one of the Southwark & Vauxhall or Lambeth Water Companies before and after the change in water source for the Lambeth Water Company. Rates for all time points were calculated based on 1851 population census

Water supply	Cholera deaths, 1849, rate per 100,000	Cholera deaths, 1854, rate per 100,000	Difference in rates comparing 1854 to 1849, rate per 100,000
Southwark & Vauxhall Company only	1349	1466	118
Lambeth Company Only	847	193	− 653
Difference-in-difference, Lambeth versus Southwark & Vauxhall	502	1273	− 771

An Estimator for the Causal Effect of Interest

An important challenge when estimating the effect of moving the water supply is that the rates of cholera death fluctuate over time. For example, the rate of cholera death increased from 1349 deaths per 100,000 people in 1849 to 1466 deaths per 100,000 in 1854 among households served by the Southwark & Vauxhall Company. Thus, if we chose to compare the rates of cholera deaths in areas supplied by the two companies at only a single time point, our resulting estimate could depend on the current distribution of cholera deaths. However, an important assumption of our causal estimand of interest is that the effect of moving the water supply is time-invariant [6•]. That is, no matter when the Lambeth Company had chosen to change its water source, the 5-year difference in cholera death rates would have been $\Theta_{\Delta t}$.

This is where the differences-in-difference estimator can help: by comparing the change in cholera death rate over time in areas served by the Lambeth Company with the change over time in areas served by the Southwark & Vauxhall Company, we can remove the time-trend from our estimate. To do so, we assume that the time-invariant effect of changing the water source can be decomposed in to an effect of time on cholera and an effect of water source on cholera:

$$E\left[\Delta t^{a=1}|A=1\right]-E\left[\Delta t^{a=0}|A=1\right]=\left(E\left[Y_{t=1}^{a=1}|A=1\right]-E\left[Y_{t=0}^{a=1}|A=1\right]\right)$$
$$-\left(E\left[Y_{t=1}^{a=0}|A=1\right]-E\left[Y_{t=0}^{a=0}|A=1\right]\right)$$

Using a set of causal inference assumptions described below, we then equate this counterfactual contrast with the observed contrast (Box 2b):

$$(E[Y|A=1,T=1]-E[Y|A=1,T=0]) - (E[Y|A=0,T=1]-E[Y|A=0,T=0]) \quad (2)$$

Finally, we use the observed rates of cholera death in 1849 and in 1854 to estimate the 5-year difference in cholera death rates (Box 2c). Figure 2 shows a graphical representation of the components of the estimator $\Theta_{\Delta t}$ [4•].

Estimating the Causal Effect of Interest

As with all causal inference methods, we cannot observe both parts of the counterfactual contrast in Eq. (1) directly. However, we may be able to get an estimate of the target difference using observed data under some reasonably plausible assumptions. The estimate given in Eq. (2) in Box 2b is the classic difference-in-difference estimate. If our scenario fits the required causal inference assumptions, then we can interpret this as an expected 5-year decrease in cholera deaths of 770.8 per 100,000 persons in the Lambeth Company customers due to the Company having chosen to move their water supply upstream.

In this simple example, we are able to hand calculate an estimate of the causal effect of interest. However, in other settings, it may be more appropriate to use a regression model to obtain the estimate. A simple regression model of the form $E[Y]=\beta_0+\beta_1(Year)+\beta_2(Company)+\beta_3(Year)\times(Company)$ can be used to provide an estimate of the same causal estimand and estimator. In this model, the coefficient on the interaction term, β_3, is the difference-in-difference estimate. A binomial or modified Poisson error distribution will typically be appropriate [7•, 8•]. Ninety-five percent confidence intervals around β_3 should be obtained using robust standard errors accounting for clustering within groups where appropriate (e.g., using SAS PROC GENMOD) [9•, 10•].

We next describe the assumptions necessary for the estimator in Eq. (2) to provide a valid estimate of causal estimand of interest in Eq. (1) and how we can use observed data to obtain each quantity given in the estimator above.

Assumptions Necessary to Endow the Estimate with a Causal Interpretation

In addition to assumptions about the quality of data and the statistical methods used, for example no measurement error or misclassification and an appropriate choice of regression model, the difference-in-difference method requires a number of assumptions to endow the final estimate with a causal interpretation (Box 3).

Box 3 Assumptions required for causal interpretation

Assumption	General example	Cholera example	Used to obtain:	Relevance to the DAG
Consistency	The observed exposure of interest is equal to the counterfactual exposure of interest	Moving the water source from opposite Hungerford Market to Thames Ditton maps to a sufficiently well-defined intervention	$E\left[Y_{t=1}^{a=1} \mid A=1\right]=E[Y \mid A=1, T=1]$ $E\left[Y_{t=0}^{a=1} \mid A=1\right]=E[Y \mid A=1, T=0]$	---
Exchangeability and parallel trends	Any unmeasured determinants of the outcome are either time-invariant or group-invariant	Any unmeasured determinants of the cholera death rate either do not vary with time or do not vary by district.	$E\left[Y_{t=1}^{a=0} \mid A=1\right]-E\left[Y_{t=0}^{a=0} \mid A=1\right]=E[Y \mid A=0, T=1]-E[Y \mid A=0, T=0]$	No common causes of A and Δt
Exchangeability and strict exogeneity	Implementation of the change in exposure was not related to the baseline value of the outcome variable	The decisions to move or not move the Lambeth Company and the Southwalk & Vauxhall Company water intake sources were independent of the cholera mortality rates in neighborhoods supplied by those companies.	--	No arrow from $Y_{t=0}$ to A No common causes of $Y_{t=0}$ and A
Positivity	All individuals or subgroups of individuals are eligible to receive all levels of exposure	The decision to move or not move the water intake source was available to both the Lambeth and Southwalk & Vauxhall Companies.	--	---

Assumption 1: Consistency

The consistency assumption requires that there exist no meaningful differences between the observed exposure of interest (moving the water source upstream) and the counterfactual exposure of interest. Though often difficult to assess, causal consistency is generally satisfied if we can describe a well-defined intervention that reflects the way in which the exposure changed in our data [11]. In the case of John Snow's exploration, changing the water source from opposite Hungerford Market to Thames Ditton seems sufficiently well-defined. We should note that this exposure is a proxy for Snow's true exposure of interest—the presence or absence of a waterborne cause of cholera. Given modern understandings of the mode of transmission of cholera, it seems likely that the change in water source is also a sufficiently well-defined intervention for the presence or absence of cholera, but in Snow's time, this was a strong, and contentious, assumption.

Consistency is violated when there are multiple ways in which an exposure can be applied. For example, it might not

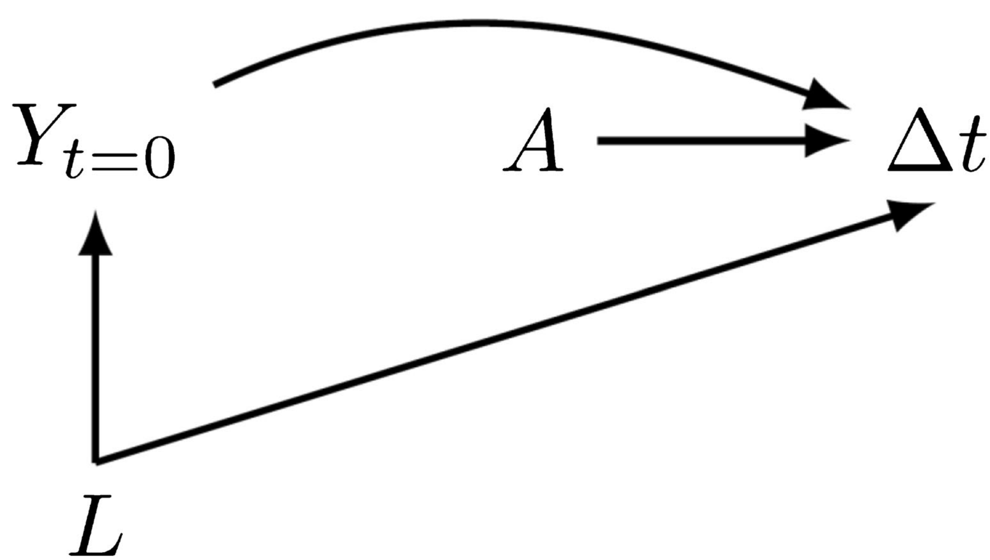

Fig. 1 Causal diagram illustrating the effect of moving the water source upstream on the change in cholera death rate from time $t=0$ to time $t=1$. Δt denotes the difference $Y_{t=1}-Y_{t=0}$

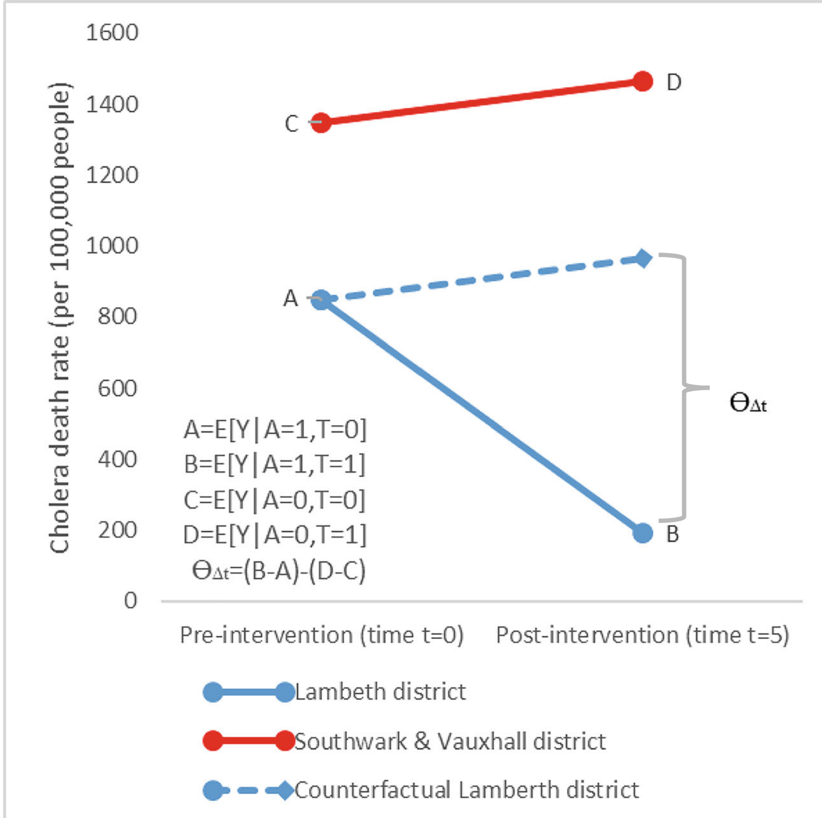

Fig. 2 A graphical representation of the components of the difference-in-difference estimator. The solid red line represents the observed cholera death rate (per 100,000 people) over time in the Southwark & Lambeth households; the solid blue line represents the observed cholera death rate (per 100,000 people) over time in the Lambeth households; and the dashed blue line represents the estimated counterfactual cholera death rate (per 100,000 people) over time in the Lambeth households had Lambeth not changed its water source. Adapted from [4•]

be appropriate to consider an exposure such as "implementation of water quality regulation" because the types of regulations implemented in various areas may differ in how much they impact cholera rates. Instead, consistency would require focusing on the implementation of a specific water quality regulation.

If the consistency assumption is met, we can directly observe the counterfactual cholera death rate for two of the four quantities we need in order to obtain the estimate of the causal effect of interest (given in Eq. (2)). First, we directly observe the cholera death rate at time $t = 0$ in the Lambeth district before the water source is changed, so $E\left[Y^{a=1}_{t=0}|A=1\right] = E[Y|A=1, T=0]$. Second, we directly observe the cholera death rate at time $t = 1$ in the Lambeth district after the water source is changed, so $E\left[Y^{a=1}_{t=1}|A=1\right] = E[Y|A=1, T=1]$.

Since the intervention occurs after time $t = 0$ and therefore cannot affect the outcome at time $t = 0$ (assuming that preparation for implementing the intervention did not affect the outcome at baseline), a version of the consistency assumption could also be used to obtain the quantity $E\left[Y^{a=0}_{t=0}|A=1\right] = E[Y|A=1, T=0]$. However, the consistency assumption is not enough to obtain the fourth quantity needed to obtain the causal effect of interest—the cholera death rate that would have been observed in the Lambeth districts at time $t = 1$ in the absence of the intervention, $E\left[Y^{a=0}_{t=1}|A=1\right]$.

Assumption 2: Exchangeability and the Parallel Trend Assumption

Two natural choices for estimating the cholera death rate that would have been observed in the Lambeth districts, if the water source had not moved upstream, are the observed cholera death rate at time $t = 1$ in the Southwark & Vauxhall districts, $E[Y| A = 0, T = 1]$, and the observed cholera death rate at time $t = 0$ in the Lambeth districts, $E[Y| A = 1, T = 0]$. However, both of these choices rely on a strong exchangeability assumption that is unlikely to be met—either that the cholera death rate in the Lambeth districts at time $t = 1$ in the absence of intervention would be the same as the observed cholera death rate at time $t = 1$ in the Southwark and Vauxhall districts, or that the cholera death rate in the Lambeth districts would not change from time $t = 0$ to time $t = 1$ in the absence of intervention.

Instead, the difference-in-difference method relies on a somewhat weaker exchangeability assumption known as the parallel, or common, trend assumption [4•]. This parallel trend assumption is usually described as the assumption that any unmeasured determinants of the cholera death rate either do not vary with time or do not vary between the districts. The parallel trend assumption leverages an assumption that *even if* the cholera death rate in the Southwark & Vauxhall districts at time $t = 1$ is not equal to the rate that would have been observed in the Lambeth districts at time $t = 1$ had they not changed their water source, the *change over time* in the cholera rates in the Southwark & Vauxhall district *is* equal to the change over time that would have been observed in the Lambeth districts in the absence of a change in water source. That is, instead of assuming that the counterfactual outcome at time $t = 1$ is equal to the observed value, we can assume that the *counterfactual difference* is equal to the *observed difference,* $E\left[Y_{t=1}^{a=0}|A=1\right] - E\left[Y_{t=0}^{a=0}|A=1\right] = E[Y|A=0,T=1] - E[Y|A=0,T=0]$.

Formally, the parallel trends assumption requires independence between A and Δt^{a}.This independence allows us to equate $E\left[Y_{t=1}^{a=0}|A=1\right] - E\left[Y_{t=0}^{a=0}|A=1\right]$ with $E\left[Y_{t=1}^{a=0}|A=0\right] - E\left[Y_{t=0}^{a=0}|A=0\right]$. Then, using consistency, we have that $\left[Y_{t=1}^{a=0}|A=0\right] - E\left[Y_{t=0}^{a=0}|A=0\right] = E[Y| A=0, T=1] - E[Y| A=0, T=0]$. Graphically, we can see that the parallel trend assumption requires that there are no common causes of A and Δt because any such common causes would vary with time and district and create the potential for bias. In contrast, common causes of $Y_{t=0}$ and Δt that vary with time but not with district (L in the DAG) do not invalidate this assumption.

We also see the parallel trend assumption at work in Fig. 2, depicted by the parallel lines illustrating the cholera death rates for the Southwark & Vauxhall district and the cholera death rates for the *counterfactual* Lambeth district. Snow justifies the parallel trend version of exchangeability between the Southwark & Vauxhall and Lambeth Companies with a detailed comparison of cholera rates in households and neighborhoods served by these companies in 1832 and 1849, before the change in water source [1•, 2•].

In practice, the parallel trend assumption cannot be empirically tested directly, since we cannot know the unobserved counterfactuals. Instead, an indirect assessment of the parallel trends is usually achieved by looking at pre-baseline trends in the outcome of interest from two or more additional time points for each group. Many other assessments and sensitivity analyses also exist to evaluate the parallel trend assumption. The parallel trend assumption will also be violated whenever factors which affect the outcome change concurrently with the change in the exposure of interest, but in only one of the groups being assessed and the existence of an outcome change of this sort can be explored as a way to evaluate the parallel trend assumption [3•]. When a concurrent change happens in *both* groups, the parallel trend assumption is not violated but consistency may no longer hold—the resulting estimate is likely only interpretable as the *joint* effect of the way in which both interventions were applied to the exposure group of interest. When the parallel trend assumption is violated, researchers might consider restricting the study question and analysis to a subset of groups or time periods in which the assumption does hold, or using weighting techniques to form synthetic control groups [3•, 12•].

A final comment about the parallel trend assumption is that in order for this assumption to be reasonable, the baseline values of the outcome of interest should be relatively similar—although not identical—between the two groups. In Snow's study, the two companies differed somewhat in their 1849 cholera death rates, which may call into question the appropriateness of the Southwark & Vauxhall Company as a control. However, the fact that the two companies served the same neighborhoods and had very similar initial water intake sources, and similar trends in cholera death rates before 1849, makes the use of the Southwark & Vauxhall Company seem reasonable. This challenge would be avoided in an individual-level randomized clinical trial, a special case of the difference-in-difference method, since the mortality rate prior to the change in water source, would be equal in expectation in the two sets of households.

Assumption 3: Exchangeability and Strict Exogeneity

The difference-in-difference literature often discusses another version of the exchangeability assumption known as strict exogeneity. Strict exogeneity requires exchangeability with respect to pre-intervention outcome levels $Y_{t=0}$. That is, the choice to intervene must not be determined by the level of the outcome at baseline. Graphically, this assumption requires that $Y_{t=0}$ and A are independent, that is, that $Y_{t=0}$ does not affect A and that there are no common causes of $Y_{t=0}$ and A.

Snow notes that the Lambeth Company moved its water intake source during a period in which there were no cases of cholera in all of London [1•, 2•]. Thus, it seems reasonable to assume that the decisions to move the source of the Lambeth Company water, and not to move the source of the Southwark & Vauxhall Company water, were made based on factors other than cholera death rates in 1849.

The strict exogeneity assumption is in fact a necessary (but not sufficient) condition for the parallel trends assumption. That is, the parallel trend assumption requires independence between A and Δt^{a}, which implies that $Y_{t=0}$ and A are also independent. If $Y_{t=0}$ and A were not independent, $Y_{t=0}$ would be a common cause of A and Δt, violating the parallel trend assumption.

Assumption 4: Positivity

All causal effect estimates require that individuals in the target population were in fact eligible to receive all levels of the exposure. In the context of a randomized trial, positivity is ensured through the assignment mechanism. In an observational study, positivity needs to be assessed by checking for any important subgroups of interest who are either ineligible to receive one level of the exposure (a structural positivity violation) or by chance happen to not receive one level of the exposure (a random positivity violation) [13]. In the context of a difference-in-difference estimator, positivity requires that both groups of interest *could have* been the group to actually experience a change in exposure level.

For Snow's example, positivity holds if it is reasonable to assume that there were no barriers preventing the Southwalk & Vauxhall Company from moving their water source, and no requirements forcing the Lambeth company to move their water source. This seems likely to be a reasonable assumption, especially since the change in water source is not required to have occurred in a particular year. Indeed, the two companies appear to have been fierce competitors—Snow reports that when a legal clause that required the Southwalk Company not to lay pipes within 2 miles of the Lambeth Company pipes was repealed in 1834, the Southwalk Company rapidly expanded into neighborhood-level competition with the Lambeth Company [1•, 2•].

Conclusions

Let us now revisit John Snow's data. Using Eq. (2), we obtain the estimate that changing the water source to a location upstream of the sewage outflow was associated with a decrease of approximately 770 deaths from cholera per 100,000 residents[1]: $\widehat{\Theta}_{\Delta t} = \left(\frac{37}{19133} - \frac{162}{19133}\right) - \left(\frac{2458}{167654} - \frac{2261}{167654}\right)$. Assuming that the four assumptions in Box 3 hold, we can interpret this estimate causally: the decision to move the Lambeth Company water intake source upstream prevented 770 deaths per 100,000 residents of households served by the Lambeth Company compared with if the Lambeth Company had not moved its water source during the time period 1849 to 1854.

This estimate tells us about the average causal effect among the treated of the *group-level* policy change in water source from downstream to upstream. We cannot necessarily conclude that moving the Southwark & Vauxhall Company water intake would result in this same reduction in mortality. Nor can we assume that the average cholera mortality in all of London would be reduced by this amount were all the water intake sources moved upstream. Generalizing beyond the target population of inference (households served by the Lambeth Company) requires additional assessment of assumptions for transportability.

Group level processes are often the intervention of interest when evaluating the impact of policies to improve population health. While much epidemiological research focuses on individual level inference, analysis of group level data rather than individual level data is appropriate when interventions are implemented at the population level. For example, analyses of cluster randomized trials focus on the impact of a group level intervention on a group level outcome. Any analysis of group level data to inform individual level understanding runs the risk of committing the ecological fallacy whereas using individual level data to infer group level causal effects could be subject to the aggregation fallacy [5].

Any attempt to interpret difference-in-difference results to make inferences about individual level processes requires understanding about the underlying mechanisms that may operate at the individual level. In Snow's case, his ultimate goal was to understand the causes of cholera at the individual level. His hypothesis was that cholera occurred via exposure to a waterborne infectious agent which in turn entered the water supply from the feces of infected individuals. Based on this hypothesis, he predicted that there would be an observable causal effect on cholera mortality rates of group level changes to water source—the original downstream water source was contaminated by London's sewage outflow, while the upstream source was not. The results of his investigation aligned with his prior hypothesis and strengthened his belief in the role of microorganisms in cholera infection. However, another scientist could have hypothesized that pipes from the Southwark & Vauxhall Company emitted a miasma that sickened households served by those companies. Such a hypothesis could also be supported by Snow's data. Neither hypothesis is directly assessed by the difference-in-difference estimator.

The difference-in-difference method thus provides us with a tool for learning about group level processes directly when these are of interest, and/or for triangulating knowledge to learn about individual level processes when individual level data are hard to obtain. The difference-in-difference method is an appropriate choice for these questions when used carefully. Importantly, the assumptions for causal inference must all be carefully assessed and determined to be appropriate. In particular, the two exchangeability assumptions can be particularly difficult to assess for difference-in-difference estimation, because of the need to assess not just confounders but how they vary with both time and exposure group.

Finally, the difference-in-difference estimate obtained from a study should only be used to describe the expected impact that the group level exposure change had *on the group that received the change.* This is an important difference from the typical causal estimands that epidemiologists are more familiar with, and therefore deserves careful and clear explanation in any difference-in-difference publication.

Although the difference-in-difference method is one of the earliest tools developed for the investigation of causal effects in observational epidemiology, it has fallen out of use given the field's shift to focus on questions of clinical relevance rather than population health relevance. However, social epidemiology and other population health-focused subfields are

[1] John Snow provides population counts for 1851, and population estimates for 1849, but no population estimates for 1854. Therefore, we have used the 1851 data for both time points.

gaining in prominence again, and attention to methods best suited to answering population level causal questions is needed. Indeed, applied examples of the use of difference-in-difference to understand social epidemiologic questions exist already [7•, 8•, 14•, 15•]. The difference-in-difference method has also been widely used in economics, and econometricians have developed a number of extensions to the simple, traditional setting we present above. These extensions are beyond the scope of this paper, but we direct interested readers to the work of Drs Susan Athey and Guido Imbens [16•], and Dr. Andrew Goodman-Bacon [17•].

The difference-in-difference method is one such method which, under strong assumptions, can provide an estimate of the causal impact of a population level exposure change on a population level outcome for the group that experienced the change. This method deserves a second look from epidemiologists interested in asking these types of causal questions.

Acknowledgments We thank Dr. Noah Haber & Dr. Samrachana Adhikari for thoughtful comments on an earlier draft of this manuscript. Any remaining errors are our own. EC was supported by NIH/NICHD K01 HD100222-01A1.

Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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